Waste Management 198 (2025) 77–86

Contents lists available at ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

Research Paper

Reducing food waste in the HORECA sector using AI-based waste-tracking devices

Evangelia G. Sigala [a], Paula Gerwin [b], Christina Chroni [a], Konstadinos Abeliotis [a], Christina Strotmann [b], Katia Lasaridi [a,*]

[a] School of Environment, Geography and Applied Economics, Harokopio University, 17671, Athens, Greece
[b] Institute of Sustainable Nutrition, Muenster University of Applied Sciences, 48149 Münster, Germany

ARTICLE INFO

Keywords:
Food waste
Foodservice sector
Waste-tracking device
Artificial intelligence
Computer vision
Waste prevention

ABSTRACT

This study assesses the effectiveness of an intervention employing an AI-based, fully automatic waste-tracking system for food waste reduction in HORECA establishments. Waste-tracking devices were installed in a restaurant within a holiday resort and a business caterer in Germany, a hotel in Switzerland, and two hotels in Greece. The devices utilize computer vision and advanced deep learning algorithms to automatically weigh and optically segregate food waste in real time. At baseline, total food waste was 76.2–121.0 g/meal for the hotels, 99.4 g/meal for the business caterer, and 151.9 g/meal for the restaurant. Avoidable food waste constituted 45 % to73% of the total, attributable to overproduction (20–92 %) and consumers' leftovers (8–80 %). The remaining waste was unavoidable, stemming from preparation procedures (47–99 %) and consumers' leftovers (1–53 %). Vegetables and prepared foods contributed the most to total amounts. This data-driven intervention raised staff awareness towards food waste, facilitating the implementation of corrective actions. Therefore, except for the Swiss hotel that exhibited an increase of 13 %, the intervention was effective in achieving a 23–51 % reduction in food waste, especially in food preparation and overproduction, demonstrating the intervention's transferability across different settings. Additional evidence supported its long-term sustainability. The cost of wasted food per meal was reduced by up to 39 % compared to the baseline. Future studies should explore combining waste-tracking devices with consumer-level interventions to enhance food waste reduction.

1. Introduction

The food waste challenge, a prominent element of unsustainable food production and consumption patterns that emerged during the Anthropocene period, undermines the resilience of global agri-food systems and impedes their transition towards sustainability (FAO, 2019; HLPE, 2014). From an environmental perspective, food waste directly imposes additional strains on planetary environmental boundaries by exacerbating the overexploitation of natural resources, excessive fertilizer applications (FAO, 2013; Kummu et al., 2012), loss of biodiversity (FAO, 2013), and greenhouse gas emissions, including methane emissions from food waste disposal in landfills (HLPE, 2014; Mbow et al., 2019; Scialabba, 2015). As such, it entails hidden costs stemming from these environmental impacts and their adverse effects on human health and well-being (FAO, 2014). Moreover, wasting food represents intrinsic financial, investment, and income losses (FAO, 2014; HLPE, 2014) and may cause a rise in food prices (HLPE, 2014). It has been associated with low labor productivity, hindered social development, social inequalities, and a higher prevalence of poverty (Cm, 2022; FAO, 2019; HLPE, 2014). Additionally, wasting food creates an ethical issue since macro- and micronutrients, as well as bioactive compounds embedded in the food matrix, are discarded amidst food insecurity, affecting approximately one-third of the world's population (FAO et al., 2024).

Although food waste is generated across the entire food supply chain, the food waste produced downstream reflects a higher loss of resources, labor, and investments, as it aggregates inputs from all the previous stages. Future projections suggest that, under a business-as-usual scenario, increasing world's population, rapid urbanization, rising per-capita affluence, and other demographic shifts driving food preferences will result in an upsurge in food demand, especially for resource-intensive foods such as meat and dairy products, necessitating a 50 % increase in agricultural production by 2050 compared to the 2012 levels (FAO, 2018). Without any intervention, food waste levels are

* Corresponding author at: Harokopio University of Athens, 17671 Greece.
E-mail address: klasaridi@hua.gr (K. Lasaridi).

https://doi.org/10.1016/j.wasman.2025.02.044
Received 30 September 2024; Received in revised form 16 February 2025; Accepted 24 February 2025
Available online 1 March 2025
0956-053X/

anticipated to exhibit an upward trend in both in- and out-of-home consumption settings. To address this issue, the United Nations (2015) has included in its Sustainable Development Targets (SDG) the target 12.3 requiring to curb food waste at the retail and consumption stages (*i. e.*, households and the foodservice sector) by 50 % by 2050. The European Commission has committed to this aspirational target (European Commission, 2015) and has since implemented several initiatives and policies to support its achievement (European Parliament and of the Council, 2018). In the forthcoming revision of the EU Waste Framework Directive, legally binding targets for food waste reduction by 30 % per-capita in retail, restaurants, food services and households, and 10 % in processing and manufacturing by 2030 are foreseen (Council of the EU, 2024; European Commission, 2023).

Approximately 1.05 billion metric tons of food were discarded globally in 2022 at the retail and consumption stages, amounting to a per-capita quantity of 132 kg (UNEP, 2024). The primary contributors were private households, accounting for 60 % of this per-capita total (79 kg/ca), followed by the foodservice sector at 28 % (36 kg/ca), and retailers at 12 % (17 kg/ca). According to the latest available statistics, in 2021 in the EU, 58.4 million tons of food (131 kg/ca) were wasted along the entire food chain, with circa 63 % arising at the consumption stages (Eurostat, 2023). Specifically, 54 % of the per-capita food waste (70 kg/ca) were discarded by householders, while 9 % (12 kg/ca) were wasted in out-of-home consumption settings.

Food waste in the foodservice sector has been characterized as an "*unsustainability hotspot*" (Eriksson et al., 2018), given its substantial impacts on social, economic, and environmental pillars of sustainability (Lins et al., 2021). This phenomenon is largely driven by an interplay of factors related to inefficiencies in operational practices (Dhir et al., 2020; Filimonau and De Coteau, 2019; Okumus, 2020), such as misestimation of expected guests, improper storage and handling (Okumus, 2020), and managerial decisions (Heikkilä et al., 2016); insufficient staff training and awareness for food waste; wasteful consumer behaviors (Dhir et al., 2020; Filimonau and De Coteau, 2019), including over-ordering (Papargyropoulou et al., 2016); undue quality standards and food safety considerations (Dhir et al., 2020; Okumus, 2020); meal purposes (Wang et al., 2017); as well as gaps in legislative frameworks and regulatory incentives (Dhir et al., 2020) that subsequently hinder the promotion of food sharing initiatives (Filimonau and De Coteau, 2019).

Food waste in out-of-home settings originates primarily from three sources: pre-kitchen or inventory-related stages, preparation and serving procedures, and plate waste left by consumers (Filimonau and De Coteau, 2019; UNEP, 2024). However, this issue is much more intricate due to the wide spectrum of foodservice types and subsectors –ranging from low turnover, privately owned businesses to multinational chain-affiliated settings, as well as establishments where food provision is not the primary or even a profit-driven activity, such as healthcare units and schools, each with different operational activity models, settings, and conditions, as well as consumer target groups, thereby resulting in diverse wasting patterns, requiring tailored reduction measures (Dhir et al., 2020; Eriksson et al., 2018). Nonetheless, it is acknowledged that the foodservice sector exhibits substantial opportunities for food waste mitigation (Eriksson et al., 2017), with companies in the sector having demonstrated the feasibility of successfully curbing food waste through targeted strategies (Beretta and Hellweg, 2019; Clowes et al., 2019, 2018a, 2018b).

Research on hotels or other short-term accommodation, restaurants, cafeterias, catering services, and canteens (HORECA), where food service constitutes a primary profit activity (*i.e.*, excluding hospitals and other healthcare facilities, educational institutions serving food, etc.), has revealed that food waste levels range between 50.1 $\pm$ 9.4 g/portion and 192.0 $\pm$ 31.0 g/portion, depending on the foodservice type (Malefors et al., 2019). In Finnish cafeterias, restaurants, and workplace canteens, food waste amounted to 102 g/portion, 153 g/portion, and 189 g/portion, respectively (Silvennoinen et al., 2015). Moreover, significant discrepancies have been noted between different restaurant types, with fine dining restaurants generating 261.1 g/day per meal served and casual dining restaurants 99.3 g/day/meal. Quick-service and limited service restaurants have typically lower food waste rates, in the range of 17 to 60 g/day/meal without considering customers' leftovers (McAdams et al., 2019). Additionally, all-you-can-eat buffets generate larger food waste amounts than à la carte services (Papargyropoulou et al., 2019), with hotels in Germany reporting a daily mean of 7.5 kg of breakfast buffet leftovers (Leverenz et al., 2021). These figures highlight the urgent need for food waste reduction initiatives in HORECA businesses. In recent years, corporate sustainability has received paramount attention (Montiel and Delgado-Ceballos, 2014), leading to the design and evaluation of numerous prevention-centered interventions.

Table 1 summarizes a range of peer-reviewed case studies that have implemented reduction interventions and initiatives in HORECA and have provided evidence regarding aspects of their impact, as defined by (Garcia-Herrero et al., 2023). The majority of existing literature has focused on consumer-targeted interventions, including nudges, *i.e.*, cues that predictably guide consumers' decision-making and choices without imposing restrictions (Garcia-Herrero et al., 2023), offering a reduction potential up to 40 % (Candeal et al., 2023). Another piece of research has highlighted the significance of staff training, increasing staff awareness, and operational changes, such as redesigning menus and decreasing portion sizes (Filimonau et al., 2017a; Filimonau and De Coteau, 2019; Montesdeoca-Calderón et al., 2024; Orr and Goossens, 2024). Additionally, mounting evidence suggests the food waste quantification process itself serves as a cornerstone of effective reduction strategies (Goossens et al., 2022). Quantification is a cornerstone in any food waste prevention effort in the food service sector: a *meta*-analysis revealed that, using different measurement methods, 61 % of catering establishments reduced their food waste in the last half of the quantification days compared to the first half of the measurement period (Eriksson et al., 2019). However, due to the inherent complexity in this sector, measuring food waste continues to present notable challenges (Eriksson et al., 2018).

Eriksson (2019) classified direct quantification methods into three groups: manual methods, semi-automatic tracking systems, and automatic tracking systems. The first category involves weighing food waste and manually recording data by staff. Semi-automatic tools feature websites or applications (apps) for data collection, but they still require staff to weigh the food waste and manually input the data. Automatic systems feature scales connected to a touchscreen electronic device (tablet or computer), where staff weigh food waste and categorize it by selecting the available menu options. Although the degree of staff involvement decreases from manual to automatic methods, all methods require additional effort and workload and thus can disrupt typical workflows (Orr and Goossens, 2024), potentially leading to underestimation depending on staff engagement and awareness. In previous research, the degree of automation in the quantification method has been positively associated with the data volume recorded (Eriksson et al., 2019). Notwithstanding differences in operation and staff involvement, companies providing semi-automatic and automatic systems differ in the level of consulting services they offer (Leverenz et al., 2021). This study expands the above classification with the additional category of fully automatic waste-tracking devices, such as KITRO, Leanpath, Lumitics, Orbisk, and Winnow which combine waste bins with software, leveraging artificial intelligence (AI) to automatically weigh and recognize the food waste items and categories minimizing staff effort. Such tools manage data and segregate waste quantities into a large number of different food items using computer vision in real time. These fully automatic, AI-based waste-tracking devices have shown effectiveness varying from 25 % to 70 % in hotels (KITRO S.A., personal communication; Leanpath, Inc., 2024; Winnow Solutions Ltd, 2024), 54 % in restaurants and bistros (Zornes, 2022), and 41 % to 50 % in workplace caterers (Leanpath, Inc., 2024; Winnow Solutions Ltd, 2024).

Table 1
Overview of studies implementing interventions for food waste prevention in HORECA.

Authors (Year)	Setting(s), country, timeframe	Methodology	Findings
Consumer-level interventions and operational changes			
Dolnicar et al. (2020)	Hotel. Slovenia. 50 days.	Game-based intervention utilizing a stamp collection booklet vs flyers asking to help reduce food waste without vs with a pro-environmental appeal vs control group.	Effectiveness of stamp collection booklet: −34 %.
Stöckli et al., (2018)	Pizzeria. Switzerland. 6 weeks.	Informational prompt about food waste in a card vs informational and normative prompt in a card to encourage boxing leftover vs control group.	Effectiveness of informational prompts: 55 % asked to box their leftovers. Effectiveness of informational and normative prompts: 64 %.
Kallbekken and Sælen (2013)	52 hotels (38 control group). Norway. 06/2012–08/2012.	Nudges to reduce plate size vs. signs with social cues encouraging guests to visit buffets more than once vs control group.	Effectiveness of plate size reduction: −19.5 %. Effectiveness of signs with cues: −20.5 %. Financial savings: NOK 50/kg/hotel (USD 9/kg/hotel). Transferability: Likely transferable to other settings with buffet-style meals. Minimal costs for implementation. No impact on guests' satisfaction.
Quantification			
(a) Semi-automatic waste-tracking devices			
Orr and Goossens (2024)	2 staff canteens. Germany. 11/2022–04/2023	Awareness-raising among kitchen staff, serving half-sized portions, providing small plates at the salad buffet, preparing soup from overproduction, offering reusable containers for takeaway, introducing meal deals at the end of serving time, and enhancing awareness of the food value.	Effectiveness: −46 % Nutritional savings: 450,000 kcal. Financial savings: €15,500. Environmental savings: 31 tons CO_2 eq. and 213 mPt PEF. Efficiency: savings of 0.03 kg of food, €2.37, 40.78 kcal, 2.74 kg CO_2 eq., and 0.02 mPt PEF per €1 invested. Transferability: High. Sustainability over time: Medium-to-high, depending on the examined measure.
(b) Automatic waste-tracking devices			
Leverenz et al. (2021) and Goossens et al. (2022)	4 hotels of a four-star hotel group. Germany. At different time points for each hotel between 2014–2015 (measurement duration: 12 months).	Quantification of breakfast buffet leftovers using a digital scale and a computer ("ResourceManager Food").	Effectiveness: −64 % Nutritional savings: 3,600,000 kcal/year/hotel. Financial savings: €8,900 per year/hotel. Environmental savings: 6.8 tons CO_2 eq. and 841 mPt PEF per year. Efficiency: savings of 0.33 kg of food, €1.69, 680 kcal, 1.30 kg CO_2 eq., and 0.16 mPt PEF per €1 invested (equipment cost: €4,500). Sustainability over time: Almost constantly low leftover levels after 5 months, although continuity depends on initial investment cost.

Abbreviation: PEF: product environmental footprint.

Up to date, there are no peer-reviewed studies on fully automatic, AI-based food waste-tracking systems, denoting a dearth of evidence-based information on the effectiveness, transferability, and long-term sustainability of such interventions. Only fragmented insights stemming from grey literature are available. Given the surging development in AI and its increasing capabilities, there is a need to elucidate the impact of AI-based interventions across different foodservice settings and geographic regions. Moreover, higher resolution measurement data in the HORECA sector remains scarce (Malefors et al., 2019). Hence, this study aims to evaluate an AI-based intervention, combining a weighing scale with computer vision in different HORECA located in Germany, Greece, and Switzerland, as part of the LOWINFOOD H2020 project.

2. Materials and methods

2.1. Study design and procedures

The LOWINFOOD research project aims to cut food waste across the whole supply chain by co-designing and implementing a portfolio of pioneering actions. By leveraging a multi-actor strategy, the project assesses the effectiveness of these actions in decreasing food waste, while also examining their broader impacts and outreach. A notable innovation investigated in this project is the utilization of AI-based waste-tracking devices in the foodservice sector, through the participation of KITRO S.A. as a project partner. KITRO S.A. is a Swiss-based company that offers a fully automatic system for data collection, real-time monitoring and analysis, as well as customized counseling for foodservice businesses. This data-driven approach enables users to optimize their operational processes, and KITRO promises their customers will experience substantial reductions in food waste, leading to substantial net cost savings.

As part of this intervention, this study employed a pre-post design comprising a baseline period followed by a demonstration stage, as illustrated in Fig. 1. At every stage, consumers were not aware of the intervention. Prior to the baseline measurements, semi-structured interviews were conducted by trained researchers with the managers of each participating establishment to assess their motives, readiness, and expectations for reducing food waste. During this preparatory phase, the managers and the KITRO team collaborated to establish the country-

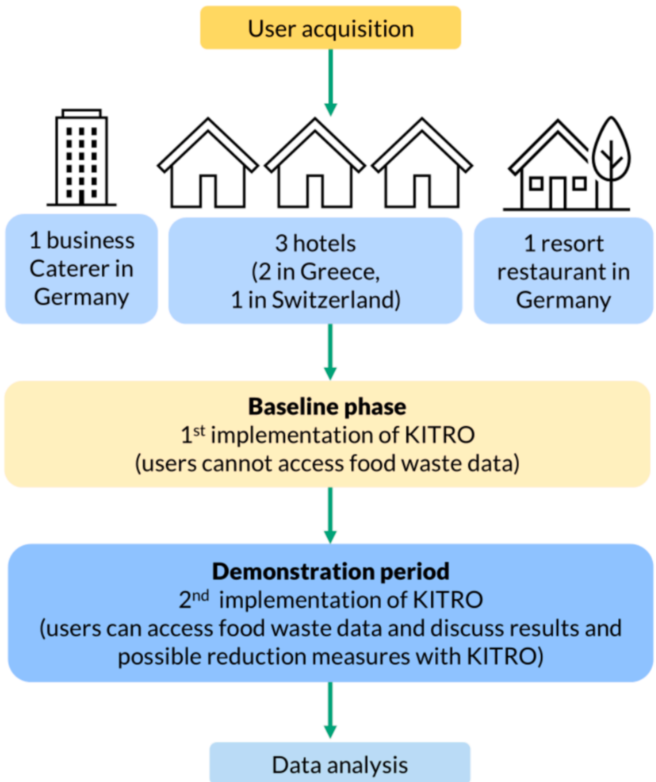

Fig. 1. Schematic illustration of the study design.

specific market prices for food items, which were then utilized in KITRO's cost calculations, with the flexibility to adjust for price changes during the whole implementation period. At the baseline stage, the waste-tracking devices were installed and measured food waste without managers having access to the collected data.

During the demonstration phase, which immediately followed the baseline, KITRO users were provided access to the online data dashboard, enabling real-time monitoring and analysis of the data recorded. Based on this information, the KITRO support team and managers held regular goal-setting meetings to discuss the findings, adjust reduction targets, and develop tailored prevention actions. After the demonstration period, follow-up semi-structured interviews were conducted with the managers to obtain insights into the accomplished goals, experiences, and difficulties encountered, as well as to assess changes induced by the implementation of the interventions.

2.2. *The waste-tracking device employed*

The waste-tracking device integrates a hardware unit comprising of a scale and an Internet of Things (IoT) device with a camera. The scale is placed underneath any type of bin and the IoT device on top of the bin. This system detects and analyzes each waste event by measuring the weight of the discarded foods with the scale, while the IoT device captures images that are sent to a cloud system, which utilizes advanced image recognition and deep learning algorithms to identify and classify food waste into food groups. Additionally, each food waste event is further segregated into specific food items. These are further characterized as avoidable and unavoidable and classified according to their source types, depending on the device location (*e.g.*, food preparation areas, kitchen, restaurants) and the algorithm. Moreover, users have access to a personalized online dashboard, providing insights into their food waste data, such as images of waste events, statistics of food waste groups, and temporal trends.

2.3. *Settings in which the waste-tracking device interventions were tested*

The users of the KITRO devices who participated in this study are described in detail in the following sections and summarized in Table 2. The primary criterion for selecting this sample was that the implementation of KITRO could yield a satisfactory return on investment in these establishments, as they handle a large number of tickets daily.

2.3.1. *Restaurant in a holiday resort in Germany, DE1*

The DE1 establishment, a restaurant located in a holiday resort in Germany, mainly served individuals renting vacation homes and other visitors. When measurements began in December 2021, the COVID-19 incidence rates in Germany were remarkably high, and thus, further restrictions were imposed, affecting also restaurants and hotels (Bundesministerium für Gesundheit, 2023). Therefore, although this restaurant usually attracts many customers during the winter season, the recorded baseline month showed the lowest guest count among all recorded months, averaging 94 guests daily.

2.3.2. *Business caterer in Germany, DE2*

DE2 was a business caterer operating in a company's office building in Germany, providing food services only for individuals associated with the company and conferences or other meetings held in the office building. Before the COVID-19 pandemic, the caterer served approximately 800 individuals daily. During the winters of 2021 and 2022, the daily guest count was reduced to nearly 150. When measurements began in May 2022, the average guest number was about 330 guests/day, but this varied significantly by weekday due to ongoing stay-at-home orders and heating energy shortages. Although the business caterer also offered some snack and breakfast items throughout the day, only the food waste generated at lunch was monitored.

2.3.3. *Hotel in Switzerland, CH*

The third KITRO user, CH, was a hotel in Switzerland with an oriental-style restaurant and a bar serving drinks and snacks. The restaurant served about 300 guests on a typical day. At the beginning of the measurements, the Swiss COVID-19 restrictions had been mostly lifted (Schweizerische Eidgenossenschaft, 2022). During the first study months, the hotel experienced a reduced overall guest count; however, the number of dinner guests remained consistent, likely due to the external customers dining at the restaurant alongside hotel guests. Over the course of the study, guest counts returned to normal. To monitor food waste from the hotel restaurant and bar, a KITRO device was installed in a hotel area. In contrast to the devices installed in other establishments, which classified food waste based on the device's location, this device utilized an advanced algorithm that considered waste mass, service time, and data from the images.

2.3.4. *Four-star all-inclusive hotel in Greece, GR1*

The fourth establishment, GR1, was a four-star all-inclusive hotel located on the island of Kos in Greece. The hotel operated on a seasonal cycle from May to October. During the 2022 and 2023 seasons, when the KITRO devices were installed, the average number of daily guests was 170 at the beginning and end of the season, while in the peak months of July and August, the average daily guest count doubled to 350.

2.3.5. *Five-star hotel in Greece, GR2*

The GR2 KITRO user was a five-star hotel on the island of Santorini in Greece. Similar to the hotel in Kos, this hotel operated on a seasonal cycle. During the implementation of the intervention in the 2022 and 2023 seasons, the guest number increased from an average of 170 guests per day in May to nearly 200–240 guests per day from July to September. By the end of the season in October, the daily guest count decreased to 100.

Table 2
Overview of foodservice settings and timeframes of the interventions.

User	DE_1	DE_2	CH	GR_1	GR_2
Country	Germany	Germany	Switzerland	Greece	Greece
Foodservice type	Resort	Business caterer	Hotel	Hotel	Hotel
Menu types					
Breakfast	Buffet (extra fee)		Buffet	Buffet	Buffet
Lunch	À la carte	Fixed menu with a self-service option	À la carte	Buffet	n.a.
Dinner	À la carte	n.a.	À la carte	Buffet	Buffet & À la carte
Number of devices	2	2	1	5	4
Baseline duration (*days*)	29	32	28	30	30
Baseline dates (*mm/yyyy*)	12/2021	05–06/2022	02/2022	05–06/2022	05/2022
Demonstration duration (*days*)	233	217	307	314	305
Demonstration dates (*mm/yyyy*)	01–11/2022	07/2022–05/2023	03/2022–01/2023	06–10/2022 & 05–10/2023	05–10/2022 & 05–10/2023
Meals count at baseline (*n*)	2,714	10,764	6,519	17,148	10,107
Meals count at demonstration (*n*)	30,845	74,387	92,342	256,885	117,107

Abbreviations: mm: month, n: absolute number, n.a.: not applicable, yyyy: year.

2.4. Food waste definitions

In this study, food waste is defined based on the European Commission's standardized terminology for the uniform food waste measurement across Member States (European Commission, 2019). In addition to once-edible food parts, the term encompasses inedible food fractions that were not segregated from the edible parts during food production.

Food waste is further classified based on its avoidability potential (WRAP, 2013a). Avoidable food waste encompasses food items or fractions that are culturally acceptable and suitable for consumption at some point before being discarded, suggesting that their disposal could have been prevented. Conversely, discarded inedible food parts are classified as unavoidable. Furthermore, food waste is categorized, based on the underlying processes or stages of its generation, into *preparation waste*, *overproduction*, and *plate waste*.

Preparation waste encompasses inedible and borderline edible food parts discarded during preparation procedures (UNEP, 2024). In this study, all preparation waste is classified as unavoidable. *Overproduction* refers to food that was discarded because it was produced in excess of demand, thus it could have been prevented. Although it may include small quantities of unavoidable waste, such as bones, here it is classified as avoidable. *Plate waste* comprises both avoidable and unavoidable consumers' leftovers, depending on their potential for prevention.

2.5. Data analyses and key performance indicators

For each establishment, the dataset included the following variables: date and time of the recorded waste events and the discarded food items, along with their food group category; weight; food waste classification based on the avoidability potential and item source; the cost for the avoidable food waste; and the device that recorded the waste event. Approximately 6–10 % of all records comprised non-food items, such as staff hands and utensils, which were removed from the datasets. Daily guest numbers were provided in a separate file by the KITRO users.

Defining effectiveness as the extent to which the interventions successfully reduced food waste levels (Garcia-Herrero et al., 2023), the impact of the interventions was assessed through a key performance indicator (KPI) calculated by dividing the summed amount of food waste generated during the baseline or demonstration phase by the total number of meals served during the respective phase:

$$\text{Per-meal food waste [g/meal]} = \frac{\sum_{i=1}^{n}(\text{Food waste amount})_i}{\sum_{i=1}^{n}(\text{Number of meals served})_i} \quad (1)$$

where $i \in \{1, 2, \ldots, n\}$ with $n \in \mathbb{N}$ representing each day of measurement during each phase. Subsequently, the percentage change from baseline was utilized to gauge the interventions' effectiveness. A similar analysis was conducted to evaluate changes in waste composition by avoidability potential, generation source and food groups, as well as cost savings from the avoidable food waste prevention.

To investigate statistical trends in food waste estimates, a second KPI was defined reflecting the amount of food waste generated on a daily basis relative to the number of meals served on that particular day, i:

$$\text{Daily per-meal food waste [g/meal/day]} = \frac{(\text{Food waste amount})_i}{(\text{Number of meals served})_i} \quad (2)$$

This KPI captures the variations in daily per-meal food waste levels, regardless of how the meal count varies from day to day.

Descriptive statistics are presented as medians (*Mdn*) and interquartile ranges, calculated as the difference between the 1st and 3rd quartiles (Q), (1st Q, 3rd Q), to represent the non-Gaussian, skewed nature of the daily per-meal food waste KPI. Differences between phases across two independent samples were evaluated utilizing the Mann-Whitney *U* test. Furthermore, the Spearman rank-order correlation was employed to explore the strength and direction of the relationship between the daily per-meal food waste quantity and the daily meal count. Statistical significance was assessed with *p*-values obtained from two-tailed hypotheses and a significance level set at 5 %. Statistical analyses were conducted using STATA version 18 (STATA Corp., College Station, Texas, USA).

Qualitative data obtained from the semi-structured interviews, including changes in practices and behaviors, was evaluated to provide further insights into the interventions' outcomes.

3. Results and discussion

3.1. Baseline total food waste levels and composition in mass and cost units

Table 3 displays the per-meal food waste during both phases. At baseline, the per-meal total food waste varied across the five users, spanning from 76.2 g/meal (GR1) to 121.0 g/meal (CH) for the hotels, 99.4 g/meal for the business caterer (DE2), and 151.9 g/meal for the restaurant in the holiday resort (DE1). This is within the range reported in the literature for similar types of facilities. A recent study reported per-portion food waste of 50.1 ± 9.4 g, 141.0 ± 6.4 g, and 192.0 ± 31.0 g for canteens, hotels, and restaurants, respectively (Malefors et al., 2019). Orr and Goossens (2024) reported baseline food waste to be 89 g/meal and 151 g/meal for two staff caterers, though excluding preparation waste. On average, 1.1 kg/guest was wasted daily in a five-star hotel in Malaysia, which offered buffet and à la carte menu (Papargyropoulou et al., 2016), while a recent review reported that HORECA businesses generate a per-meal average of 209 g of food waste (Vardopoulos et al.,

Table 3
Comparative analysis of per-meal food waste levels by avoidability potential and the stage of generation, as well as cost of per-meal avoidable food waste, across the intervention phases.

User	Baseline				Demonstration phase				Change in total levels (%)
	Operational level		Costumers		Operational level		Costumers		
	Overproduction	Preparation waste	Avoidable plate waste	Unavoidable plate waste	Overproduction	Preparation waste	Avoidable plate waste	Unavoidable plate waste	
	Mass of food waste per served meal (g/meal)								
DE1	41.5	40.6	55.0	14.7	18.4	29.3	41.9	11.8	–33 %
DE2	43.3	36.3	19.3	0.5	26.6	19.1	25.0	5.8	–23 %
CH	55.8	27.1	32.7	5.5	63.6	34.8	33.5	4.5	+13 %
GR1	6.9	19.8	27.1	22.4	4.2	3.0	32.1	14.9	−29 %
GR2	41.8	30.1	3.7	1.0	22.9	5.7	7.2	1.9	−51 %
	Cost of avoidable food waste per served meal (€/meal or CHF/meal[‡])								
DE1	0.26	n.a.	0.34	n.a.	0.08	n.a.	0.28	n.a.	−39 %
DE2	0.29	n.a.	0.11	n.a.	0.15	n.a.	0.17	n.a.	−21 %
CH	0.41	n.a.	0.24	n.a.	0.42	n.a.	0.24	n.a.	+1%
GR1	0.03	n.a.	0.14	n.a.	0.01	n.a.	0.16	n.a.	−1%
GR2	0.25	n.a.	0.03	n.a.	0.13	n.a.	0.06	n.a.	−34 %

Values are described as the sum of each variable divided by the total number of meals served.
Abbreviations: CH: hotel in Switzerland; DE1: restaurant in a holiday resort in Germany; DE2: business caterer in Germany; GR1: hotel in Greece; GR2: hotel in Greece, n.a.: not applicable.
[†] Change refers to the percentage change in per-meal total food waste from the baseline to the demonstration phase.
[‡] The per-meal cost is expressed in € for foodservices located in Germany and Greece. For the Swiss hotel, this cost is expressed as CHF/meal.

2025).

The percentage of avoidable food waste ranged significantly, from 45 % at GR1 to 73 % at CH (GR2: 59 %; DE2: 63 %; DE1: 64 %). The composition of this avoidable fraction exhibited considerable variability among the establishments. Specifically, overproduction ranged widely, from 20 % at GR1 to 92 % at GR2 (DE1: 43 %; CH: 63 %; DE2: 69 %). Moreover, the avoidable fraction of plate waste ranged from 8 % at GR2 to 80 % at GR1 (DE2: 31 %; CH: 37 %; DE1: 57 %). A similar broad range was observed in the proportion of unavoidable plate waste relative to the total unavoidable waste, spanning from 1 % at DE2 to 53 % at GR1 (GR2: 3 % CH: 17 %; DE1: 27 %). Conversely, the share of preparation waste on total unavoidable food waste was consistently high across most establishments (DE1: 73 %; CH: 83 %; GR2: 97 %; DE2: 99 %), except for GR1 (47 %). Literature data from German foodservice businesses showed that 25–55 % of food waste was due to overproduction, 25–30 % from plate waste, and 20–35 % occurred during preparation (Borstel et al., 2020). Moreover, in the UK, 75 % of food waste in hospitality businesses was classified as avoidable, with overproduction accounting for 45 %, 34 % from consumers leftovers, and 21 % from food spoilage (WRAP, 2013b). Similarly, 56 % of food discarded in a five-star hotel in Malaysia could have been prevented, while the remaining 44 % was attributed to the preparation of high-quality meals from scratch, resulting in substantial volumes of inedible food parts (Papargyropoulou et al., 2016). In another study, avoidable food waste ranged from 32 % to 63 % of total food waste (Papargyropoulou et al., 2019).

Table 4 illustrates per-meal food waste by food groups. Vegetables accounted for the largest share of total food waste per meal at GR2 (47 %) and DE1 (51 %). Prepared foods, such as scrambled eggs, French fries, salads, pizza and sandwiches, stews, etc., constituted the highest proportion of per-meal total food waste at GR1 (31 %), DE2 (53 %), and CH (54 %). However, at GR1, vegetables and fruits combined reached 49 %. Vegetables, cereals, and fruit have been identified as the top three most wasted foods, with vegetables and fruits mostly contributing to preparation waste and cereals to consumers' leftovers (Papargyropoulou et al., 2016).

In economic terms, the baseline cost of per-meal avoidable food waste ranged from 0.17 €/meal (GR1) to 0.59 €/meal (DE1), and 0.65 CHF/meal at CH (Table 3). The cost of food discarded due to overproduction varied substantially from 20 % at GR1 to 90 % at GR2 (DE1: 43 %; CH: 63 %; DE2: 72 %). Similarly, the contribution of avoidable plate waste also varied widely from 10 % at GR2 to 83 % at CH (DE2: 37 %; DE1: 57 %; GR1: 80 %). Other sources reported an economic loss of 23 % of food purchased (Papargyropoulou et al., 2019).

3.2. Effectiveness of the interventions

The installation of the waste-tracking devices led to a reduction in per-meal total food waste by 23–51 % for all establishments, except for CH, which exhibited a modest increase of 13 % (Table 3). Specifically, this rise at CH was driven by a 21 % increase in the unavoidable fraction and a 10 % increase in avoidable food waste. Conversely, significant reductions in the avoidable fraction were observed at DE2 (−38 %), GR2

Table 4
Comparative analysis of per-meal food waste levels by food groups, across the intervention phases.

Food waste category	Baseline					Demonstration phase				
	DE_1	DE_2	CH	GR_1	GR_2	DE_1	DE_2	CH	GR_1	GR_2
Vegetables (*g/meal*)	77.5	25.7	23.6	18.9	36.2	43.3	12.4	23.9	4.6	8.3
Fruits (*g/meal*)	11.6	5.4	6.3	18.6	9.4	7.9	0.7	4.0	3.3	2.0
Prepared foods (*g/meal*)	46.0	52.4	54.6	24.0	18.6	36.5	51.5	73.5	33.6	16.1
Confectionery products (*g/meal*)	4.2	0.5	18.3	4.8	2.6	3.6	1.0	15.4	4.2	3.5
Starchy foods (*g/meal*)	5.3	10.5	5.3	4.6	6.3	3.1	6.6	7.7	2.5	2.3
Dairy products (*g/meal*)	0.6	0.2	5.6	1.0	2.1	0.6	0.5	4.2	0.6	0.9
Meat & seafood (*g/meal*)	3.8	1.4	7.1	0.9	1.1	3.3	1.6	5.6	3.0	1.9
Animal-derived inedible food parts (*g/meal*)	3.0	3.2	0.1	3.4	0.4	3.1	2.2	1.9	2.4	2.7

Values are described as the sum of each variable divided by the total number of meals served.
Abbreviations: CH: hotel in Switzerland; DE1: restaurant in a holiday resort in Germany; DE2: business caterer in Germany; GR1: hotel in Greece; GR2: hotel in Greece.

(−34 %), and DE1 (−18 %). There is a dearth of studies utilizing AI-based devices for waste tracking. To the best of our knowledge, only self-reported, non-peer-reviewed results from corporate initiatives using fully automatic waste-tracking devices are available. These have reported effectiveness ranging widely, from 25-70 % in hotels (KITRO S.A., personal communication; Leanpath, Inc., 2024; Winnow Solutions Ltd, 2024), 54 % in restaurants and bistros (Zornes, 2022), and 41–50 % in workplace caterers (Leanpath, Inc., 2024; Winnow Solutions Ltd, 2024). Moreover, Orr and Goossens (2024) reported a 46 % reduction in waste in staff canteens using semi-automatic waste-tracking devices, while Leverenz et al. (2021) observed a 64 % reduction in breakfast buffet leftovers in four hotels in Germany with the implementation of non-AI-based automatic waste-tracking systems. As typically higher amounts of food waste are generated in buffets than in à la carte menu (Papargyropoulou et al., 2019), it is likely that, overall, there was a greater potential for food waste prevention in the Leverenz et al. (2021) study. Still, the results of these studies, using non-AI-based waste-tracking systems, imply that waste-tracking systems with technologies of lower complexity can achieve comparable effectiveness. However, a significant restriction of the latter systems is their requirement for higher staff involvement, making long term adherence to the recording process more demanding.

While adjustments to mitigate overproduction, based on KITRO results, were recorded for all four users (DE1: −56 %; DE2: −39 %; GR1: −39 %; GR2: −45 %) decreases in avoidable plate waste were only observed at DE1 (−24 %). The unavoidable food waste dropped for all users except CH (GR1: −76 %; GR1: −58 %; DE2: –32 %; DE1: −26 %), with declines in preparation waste being more pronounced (GR1: −85 %; GR2: −81 %; DE2: −47 %; DE1: −28 %) compared to unavoidable plate waste, which was decreased only at two users (GR1: –33 %; DE1:-20 %). Additionally, the intervention's impact on specific food waste categories revealed that fruits exhibited the largest reductions among eight food groups (DE1: −87 %; GR1: −83 %; GR2: −79 %; CH: −36 %), while vegetables showed the largest decrease at DE1 (−44 %) (Table 4).

Upon examining the effectiveness of the interventions in economic terms, cost savings from decreasing per-meal avoidable food waste ranged from 1 % to 39 % for all establishments except for CH (Table 3). Specifically, compared to baseline, the cost associated with food overproduction dropped in four establishments (DE1: −69 %; GR1: −66 %; GR2: −50 %; DE2: −48 %) during the demonstration phase, whereas the cost of plate waste decreased only at DE1 (−16 %). Based on non-peer-reviewed data, in other HORECA financial savings of 51–66 % have been reported using fully automated tracking devices (Leanpath, Inc., 2024), while in a hotel with an average of 400 guest/day, KITRO implementation resulted in 170 % return on investment after 34 months (KITRO S.A., personal communication). Leverenz et al. (2021) reported monetary savings of 65 % through the implementation of automatic waste-tracking systems in hotels. In that study, the most significant savings were seen in the categories of fruits, cheese, fish, and other items, each experiencing a decline exceeding 80 %.

Regarding the longitudinal changes in daily per-meal food waste levels, similar trends were observed across all establishments (Table 1S). Specifically, except CH, the interventions resulted in a statistically significant decrease in the median daily total food waste generated per meal from the baseline to the demonstration phase (p-values for differences: DE1: 0.007; DE2: 0.005; GR1: <0.001; GR2: <0.001). Conversely, at CH, the median daily per-meal total food waste increased (p-value for difference: 0.037). Further analysis using the Mann-Whitney U test revealed, in most cases, statistically significant changes in all four food waste categories examined, providing evidence that improvements in all respective processes were the primary underlying drivers of the observed reductions.

3.3. Qualitative outcomes and the link between food waste and meal count

Interviews with managers combined with an investigation of the correlation between food waste and meal count (Table 2S) were utilized to identify underlying patterns, and operational and behavioral changes induced by the intervention.

3.3.1. Restaurant in a holiday resort in Germany, DE1

Before the installation of waste-tracking devices, managers reported being unaware of their food waste levels; thus, the lack of significant correlations between the daily per-meal food waste and meal count during this period may reflect poor demand forecasting and large portion sizes, along with limitations in food redistribution due to COVID-19 restrictions (Table 2S).

Nonetheless, at the demonstration phase, a weak negative correlation emerged between total food waste per meal and the number of meals served daily. This suggests that the corrective measures adopted on the basis of KITRO results may have been more effective in reducing food waste as the daily meal count increased. The follow-up interview revealed that these measures included good food management practices, such as more frequent restocking of the breakfast buffet with smaller food quantities, reduction in unnecessary overproduction, and portion size adjustments to meet the actual food demand.

3.3.2. Business caterer in Germany, DE2

At the first semi-structured interview conducted before the baseline, the management of the business caterer asserted existing endeavors to reduce food waste. This is likely reflected in the negative correlations between the daily per-meal food waste and daily meal count at baseline, possibly indicating a more effective demand forecasting as the daily meal count increases (Table 2S).

At the demonstration phase, the previously strong negative correlation between the daily per-meal total food waste and the number of meals served daily weakened. This suggests that, as information on food waste generated became more available through KITRO, the impact of the food waste reduction efforts became less dependent on the number of meals served daily. Additionally, the association between the daily meal count and preparation waste lost its statistical significance, indicating that preventive measures, such as the menu adjustments stated by the managers during the follow-up interview, may have contributed to food waste reduction independently of the number of meals served daily. Conversely, the association with daily per-meal overproduction food waste surpassed the statistical significance threshold, shifting to a very weak negative correlation, implying that preventive measures targeting overproduction were more effective when economies of scale could be attained, albeit to a limited extent.

3.3.3. Hotel in Switzerland, CH

Prior to the baseline, the hotel managers revealed their motivation and existing, albeit fragmented, efforts to decrease food waste. The negative associations between the daily per-meal food waste and the daily meal count likely reflect again a more effective demand forecasting as the number of meals served daily increases (Table 2S).

During the demonstration phase, there was a slight weakening of the negative correlation between the daily meal count and the daily per-meal total food waste and unavoidable preparation waste. Nonetheless, the food waste variation between the baseline and demonstration phase has been likely influenced by external factors, such as the COVID-19 pandemic, which made food demand forecasting more challenging, especially for external dinner guests. This is reflected in the increase of overproduction and preparation food waste in the demonstration period (Table 3). During the follow-up interview, the hotel management confirmed that measures for reducing food waste, such as frequent restocking breakfast buffets and discussions with staff, were initiated, but they were not fully implemented. Furthermore, high staff turnover and a shift to a four-day workweek have impeded the implementation of the preventive actions. Additionally, according to management, some staff members exhibited unawareness towards the amount of food waste.

3.3.4. Four-star all-inclusive hotel in Greece, GR1

Before the baseline period, hotel managers declared that they had already made attempts to curb food waste. Delving deeper into the correlations, the negative correlations between daily per-meal food waste and the number of meals served daily indicate that these efforts were more effective in larger guest counts (Table 2S). At baseline, there was a moderate increase in daily avoidable plate waste per meal as the number of daily meals rose. This may be attributed to the baseline coinciding with the beginning of the summer season, during which new menu options were tested by the chefs, as reported by the hotel management. It is possible that guests did not like all these menu options, leaving more food on their plates.

At the demonstration phase, only the association between daily meal count and daily per-meal preparation waste retained its statistical significance, shifting to a weak positive relationship. This shift may be indicative of the increased availability of summer fruits on the buffets, such as watermelons and melons, which contain a high proportion of inedible parts. Moreover, the intervention's effectiveness in reducing food waste and the loss of statistical significance in other correlations suggest that re-designing the menu and staff training, as reported by hotel management, may have been effective preventive measures irrespective of how many meals were served daily.

3.3.5. Five-star hotel in Greece, GR2

Prior to the installation of waste-tracking devices, hotel management was motivated to reduce food waste. However, they noted that this objective occasionally contradicted their willingness to satisfy their guests. This may explain the strong positive associations observed at baseline between daily per-meal plate waste and daily meal count (Table 2S), as the desire of the hotel management to provide many food options and high food availability in the buffets resulted to an abundance of food presented, likely leading guests to overload their plates (Papargyropoulou et al., 2019). This was more pronounced the higher the guest count, as more food was made available in the buffets.

At the demonstration period, daily per-meal total food waste, as well as food waste due to preparation and overproduction, were negatively correlated with the number of meals served daily. This indicates that the implementation of preventive measures was more effective in high guest counts. Hotel managers reported improvements in food management practices, such as better food storage to reduce spoilage; decreased food overproduction; implementation of a corporate policy for redistributing unconsumed food from the kitchen (i.e. overproduction) to staff; portion size adjustments; an on-demand preparation of foods, like scrambled eggs, at the buffets; and staff training. Similarly to the hotel in Kos, the very weak correlation between daily per-meal unavoidable plate waste and the daily meal count may be linked to consumers' dietary preferences, such as the consumption of summer fruits with a high proportion of inedible parts, which are served with peelings at this hotel.

3.4. Assessment of the interventions, strengths and limitations

The effectiveness of the intervention in lowering food waste varied across the different settings examined, with the largest reduction observed at GR2, which implemented the greatest number of tailored measures. Therefore, systematically measuring food waste, in conjunction with other measures (Eriksson et al., 2019), provides HORECA businesses with valuable insights into the magnitude of the problem, allows for the identification of food waste hotspots, and enhances managers' and staff's awareness, thereby facilitating the implementation of targeted corrective measures (Goossens et al., 2022; Leverenz et al., 2021; Strotmann et al., 2022).

Moreover, contrary to previous findings (Eriksson et al., 2019), high baseline food waste levels do not seem to influence the effectiveness of quantification interventions (Table 3). Instead, managers' and staff's engagement, awareness, and willingness to implement the prevention measures seem to be more important. This is particularly evident at the CH, in which per-meal total food waste increased. This increase was likely due to several factors that hampered food demand forecasting and redistribution of uneaten food, such as the COVID-19 pandemic. Additional factors include limited opportunities for staff training due to the reported high staff turnover, a weak motivation of managers and staff, or possibly their resistance to change – a limitation of our study, yet reflective of real-world settings. These issues are commonly reported by other researchers (Filimonau and De Coteau, 2019; Martin-Rios et al., 2018; Montesdeoca-Calderón et al., 2024). However, the increase in food waste at CH cannot be conclusively attributed to any single factor, as these aspects were not specifically explored in our analysis, which constitutes another limitation of our study. Overall, as discussed in prior studies (Dhir et al., 2020; Malefors et al., 2019; Papargyropoulou et al., 2016), socio-cultural and geographic differences, diverse operational activities, settings, and menu types, along with the differentiated impact of external factors, like COVID-19 and heating energy shortages, may have accounted for these differences between settings. However, not all establishments were affected by these external factors, such as COVID-19 and energy shortages; only DE1, DE2, and CH experienced impacts. Specifically, DE1 had a lower guest count at baseline due to COVID-19 and energy shortages; in DE2, the intervention started in May 2022, after COVID-19, with a lower guest count; and in CH, although Swiss COVID-19 restrictions had been mostly lifted, the guest count remained lower than usual. Therefore, with relatively higher guest counts during the demonstration phases, this could have undermined the results of the interventions. Moreover, another limitation is that the possibility of the Hawthorne effect, behavioral reactivity (Jones, 1992), and social desirability bias induced by the self-observation process, as well as misclassification errors or incomplete recording of waste events in the waste-tracking device cannot be entirely dismissed, particularly in the CH, where staff motivation was low and only one KITRO device was available. The results from KITRO devices were not validated through on-site waste compositional analyses by a researcher or a third party. However, images uploaded to the dashboard were thoroughly checked by the research team through random sampling.

A key strength of this study is the successful implementation of the intervention across different contextual and geographic settings, which demonstrates its transferability. However, the cost of the initial investment cannot be assessed, as the interventions were part of the LOWINFOOD project. Thus, the financial efficiency of the intervention cannot be evaluated. However, follow-up communication with the managers of GR2 a year later confirmed that they decided to invest in the KITRO technology, when this study ended. Moreover, the company owing DE1 decided to install KITRO devices in other establishments it owns. This suggests that the companies considered the intervention to be useful to sustain good food waste prevention practices over time and cost-effective for their overall business priorities.

A practical asset, highlighted by all establishments involved in this study, was the user-friendly online dashboard and the automatically produced reports for managers and chefs. All users of KITRO reported that these outputs provided clear daily understanding of the food waste hot spots, required minimal effort and time to assess related performance indicators, and were cost-effective considering the final savings. Moreover, the high data granularity provided without additional staff workload facilitated the detailed detection of food waste sources.

This intervention type was more effective in reducing preparation food waste and overproduction than consumers' leftovers. Therefore, it may be more suitable for targeting food waste related to operational procedures and demand forecasting. This underscores the need to also address consumers' plate waste by combining behavioral interventions that raise consumers' awareness, such as nudges, along with interventions using waste-tracking devices, thereby facilitating food waste reduction at all levels of foodservice. This combined approach could also amplify the potential positive systemic effects of waste-tracking devices, such as when trained staff adopt pro-environmental behaviors at their households (Wang et al., 2022).

4. Conclusions

This study is unique, as it provides both quantitative and qualitative evidence on the results attained by interventions using AI-based, fully automatic food waste tracking devices, to facilitate and guide the development and implementation of tailored prevention measures at the operational level. The installation of these devices resulted in a 23–51 % reduction in per-meal total food waste across different settings, with reported positive evidence of sustainability over time. In these establishments, the intervention enhanced staff awareness towards food waste, facilitating prevention actions, at the operational level, such as mitigation of food overproduction, regular buffet restocking, portion and menu adjustments, food storage improvement, redistribution of uneaten food to staff, on-demand food preparation at buffets, and staff training. Conversely, a 13 % increase in food waste observed in one of the establishments examined, was mainly attributed to external factors, such as the COVID-19 pandemic, and low motivation of managers and staff. Future studies should focus on combined interventions that address both operational and consumer level, while government and legislative bodies should support incentives for implementing these projects.

CRediT authorship contribution statement

Evangelia G. Sigala: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Paula Gerwin:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christina Chroni:** Writing – review & editing, Project administration, Investigation, Data curation, Conceptualization. **Konstadinos Abeliotis:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Christina Strotmann:** Writing – review & editing, Project administration, Investigation, Data curation, Conceptualization. **Katia Lasaridi:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Funding

LOWINFOOD has received funding from the European Union's Horizon 2020 Research and Innovation programme under grant agreement No 101000439.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Authors would like to thank KITRO S.A. for their cooperation during the demonstration phase. The trusting collaboration and the provision of internal data formed the basis for our study. We would also like to thank the participating establishments, in which KITRO devices were installed, for their support in providing the required data for our analyses. Businesses' cooperation for data publishing is never a matter of course.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2025.02.044.

Data availability

Data will be made available on request.

References

Beretta, C., Hellweg, S., 2019. Potential environmental benefits from food waste prevention in the food service sector. Resour. Conserv. Recycl. 147, 169–178. https://doi.org/10.1016/j.resconrec.2019.03.023.

Bundesministerium für Gesundheit, 2023. Coronavirus-Pandemie: Was geschah wann? [WWW Document]. BMG. URL https://www.bundesgesundheitsministerium.de/coronavirus/chronik-coronavirus (accessed 9.2.24).

Candeal, T., Brüggemann, N., Bruns, H., Casonato, C., Diercxsens, C., Garcia Herrero, L., Gil, J.M., Haglund, Y., Kaptan, G., Kasza, G., Mikkelsen, B.E., Obersteiner, G., Pires, I.M., Swannell, R., Vainioranta, J., Van Herpen, E., Vittuari, M., Watanabe, K., Sala, S., 2023. Tools, best practices and recommendations to reduce consumer food waste - A compendium. Publications Office of the European Union, Luxembourg, JRC133004. doi: 10.2760/404597.

Clowes, A., Mitchell, P., Hanson, C., 2018a. The Business Case for Reducing Food Loss and Waste: Catering, The Business Case for Reducing Food Loss and Waste.

Clowes, A., Mitchell, P., Hanson, C., 2018b. The Business Case for Reducing Food Loss and Waste: Hotels, The Business Case for Reducing Food Loss and Waste.

Clowes, A., Hanson, C., Swannell, R., 2019. The Business Case for Reducing Food Loss and Waste: Restaurants, The Business Case for Reducing Food Loss and Waste.

Cm, J., 2022. Impacts of food wastage on economic growth. World Food Policy 8, 118–125. https://doi.org/10.1002/wfp2.12038.

Council of the EU, 2024. Waste framework directive: Council set to start talks on its revision [WWW Document]. Consilium. URL https://www.consilium.europa.eu/en/press/press-releases/2024/06/17/waste-framework-directive-council-set-to-start-talks-on-its-revision/ (accessed 7.20.24).

Dhir, A., Talwar, S., Kaur, P., Malibari, A., 2020. Food waste in hospitality and food services: A systematic literature review and framework development approach. J. Clean. Prod. 270, 122861. https://doi.org/10.1016/j.jclepro.2020.122861.

Dolnicar, S., Juvan, E., Grün, B., 2020. Reducing the plate waste of families at hotel buffets – A quasi-experimental field study. Tour. Manage. 80. https://doi.org/10.1016/j.tourman.2020.104103.

Eriksson, M., Persson Osowski, C., Malefors, C., Björkman, J., Eriksson, E., 2017. Quantification of food waste in public catering services – A case study from a Swedish municipality. Waste Manag. 61, 415–422. https://doi.org/10.1016/j.wasman.2017.01.035.

Eriksson, M., Persson Osowski, C., Björkman, J., Hansson, E., Malefors, C., Eriksson, E., Ghosh, R., 2018. The tree structure — A general framework for food waste quantification in food services. Resour. Conserv. Recycl. 130, 140–151. https://doi.org/10.1016/j.resconrec.2017.11.030.

Eriksson, M., Malefors, C., Callewaert, P., Hartikainen, H., Pietiläinen, O., Strid, I., 2019. What gets measured gets managed – Or does it? Connection between food waste quantification and food waste reduction in the hospitality sector. Resour. Conserv. Recycl.: X 4. https://doi.org/10.1016/j.rcrx.2019.100021.

European Commission, 2015. Communication from the Commission to the European Parliament, The Council, The European Economic and Social Committee and The Committee of the Regions Closing the loop - An EU action plan for the Circular Economy, COM(2015) 614 final.

European Commission, 2019. Commission Delegated Decision (EU) 2019/1597 of 3 May 2019 supplementing Directive 2008/98/EC of the European Parliament and of the Council as regards a common methodology and minimum quality requirements for the uniform measurement of levels of food waste, (OJ L 248, 27.9.2019, p. 77–85).

European Commission, 2023. Proposal for a Directive of the European Parliament and of the Council amending Directive 2008/98/EC on waste, COM (2023) 420 final.

European Parliament and of the Council, 2018. Directive (EU) 2018/851 of the European Parliament and of the Council of 30 May 2018 amending Directive 2008/98/EC on waste, (OJ L 150, 14.6.2018, p. 109–140).

Eurostat, 2023. Food waste and food waste prevention by NACE Rev. 2 activity - tonnes of fresh mass. URL https://ec.europa.eu/eurostat/databrowser/view/env_wasfw/default/table?lang=en (accessed 11.8.23).

FAO, 2014. Food Wastage Footprint: Full-Cost Accounting (Final Report). FAO, Rome, Italy.

FAO, IFAD, UNICEF, WFP, WHO, 2024. The State of Food Security and Nutrition in the World 2024: Financing to end hunger, food insecurity and malnutrition in all its forms. FAO, IFAD, UNICEF, WFP, WHO, Rome, Italy.

FAO, 2013. Food wastage footprint: Impacts on Natural Resources. Summary Report. FAO, Rome, Italy.

FAO, 2018. The future of food and agriculture-Alternative pathways to 2050. FAO.

FAO, 2019. The State of Food and Agriculture 2019: Moving Forward on Food Loss and Waste Reduction. FAO. doi: 10.18356/32f21f8c-en.

Filimonau, V., De Coteau, D.A., 2019. Food waste management in hospitality operations: A critical review. Tour. Manage. 71, 234–245. https://doi.org/10.1016/j.tourman.2018.10.009.

Filimonau, V., Lemmer, C., Marshall, D., Bejjani, G., 2017a. Restaurant menu re-design as a facilitator of more responsible consumer choice: An exploratory and preliminary study. J. Hosp. Tour. Manag. 33, 73–81. https://doi.org/10.1016/j.jhtm.2017.09.005.

Garcia-Herrero, L., Casonato, C., Caldeira, C., De Laurentiis, V., Bruns, H., Sala, S., 2023. Scoping consumer food waste: an evaluation framework of prevention interventions.

Publications Office of the European Union, Luxembourg, JRC128763. doi: 10.2760/3128.
Goossens, Y., Leverenz, D., Kuntscher, M., 2022. Waste-tracking tools: A business case for more sustainable and resource efficient food services. Resour. Conserv. Recycl. 15, 200112. https://doi.org/10.1016/j.rcradv.2022.200112.
Heikkilä, L., Reinikainen, A., Katajajuuri, J.-M., Silvennoinen, K., Hartikainen, H., 2016. Elements affecting food waste in the food service sector. Waste Manag. 56, 446–453. https://doi.org/10.1016/j.wasman.2016.06.019.
Hlpe, 2014. Food losses and waste in the context of sustainable food systems. A report by the High Level Panel of Experts on Food Security and Nutrition of the Committee on World Food Security. HPLE, Rome, Italy.
Jones, S.R.G., 1992. Was There a Hawthorne Effect? Am. J. Sociol. 98, 451–468.
Kallbekken, S., Sælen, H., 2013. 'Nudging' hotel guests to reduce food waste as a win–win environmental measure. Econ. Lett. 119, 325–327.
Kummu, M., de Moel, H., Porkka, M., Siebert, S., Varis, O., Ward, P.J., 2012. Lost food, wasted resources: Global food supply chain losses and their impacts on freshwater, cropland, and fertiliser use. Sci. Total Environ. 438, 477–489. https://doi.org/10.1016/j.scitotenv.2012.08.092.
Leanpath, Inc., 2024. Resources - Leanpath [WWW Document]. URL https://www.leanpath.com/resources/#case (accessed 9.10.24).
Leverenz, D., Hafner, G., Moussawel, S., Kranert, M., Goossens, Y., Schmidt, T., 2021. Reducing food waste in hotel kitchens based on self-reported data. Ind. Mark. Manag. 93, 617–627.
Lins, M., Puppin Zandonadi, R., Raposo, A., Ginani, V.C., 2021. Food waste on foodservice: an overview through the perspective of sustainable dimensions. Foods 10. https://doi.org/10.3390/foods10061175.
Malefors, C., Callewaert, P., Hansson, P.-A., Hartikainen, H., Pietiläinen, O., Strid, I., Strotmann, C., Eriksson, M., 2019. Towards a baseline for food-waste quantification in the hospitality sector—quantities and data processing criteria. Sustainability 11. https://doi.org/10.3390/su11133541.
Martin-Rios, C., Demen-Meier, C., Gössling, S., Cornuz, C., 2018. Food waste management innovations in the foodservice industry. Waste Manag. 79, 196–206. https://doi.org/10.1016/j.wasman.2018.07.033.
Mbow, C., Rosenzweig, C., Barioni, L.G., Benton, T.G., Herrero, M., Krishnapillai, M., Liwenga, E., Pradhan, P., Rivera-Ferre, M.G., Sapkota, T., Tubiello, F.N., Xu, Y., 2019. Food Security, in: Climate Change and Land: An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems [P.R. Shukla, J. Skea, E. Calvo Buendia, V. Masson-Delmotte, H.-O. Pörtner, D.C. Roberts, P. Zhai, R. Slade, S. Connors, R. van Diemen, M. Ferrat, E. Haughey, S. Luz, S. Neogi, M. Pathak, J. Petzold, J. Portugal Pereira, P. Vyas, E. Huntley, K. Kissick, M. Belkacemi, J. Malley, (Eds.)].
McAdams, B., von Massow, M., Gallant, M., Hayhoe, M.-A., 2019. A cross industry evaluation of food waste in restaurants. J. Foodserv. Bus. Res. 22, 449–466. https://doi.org/10.1080/15378020.2019.1637220.
Montesdeoca-Calderón, M.-G., Gil-Saura, I., Ruiz-Molina, M.-E., Martin-Rios, C., 2024. Tackling food waste management: Professional training in the public interest. Int. J. Gastron. Food Sci. 35, 100863. https://doi.org/10.1016/j.ijgfs.2023.100863.
Montiel, I., Delgado-Ceballos, J., 2014. Defining and measuring corporate sustainability: are we there yet? Organ. Environ. 27, 113–139. https://doi.org/10.1177/1086026614526413.
Okumus, B., 2020. How do hotels manage food waste? evidence from hotels in Orlando, Florida. J. Hosp. Mark. Manage. 29, 291–309. https://doi.org/10.1080/19368623.2019.1618775.
Orr, L., Goossens, Y., 2024. Trimming the plate: a comprehensive case study on effective food waste reduction strategies in corporate canteens. Sustainability 16, 785. https://doi.org/10.3390/su16020785.
Papargyropoulou, E., Wright, N., Lozano, R., Steinberger, J., Padfield, R., Ujang, Z., 2016. Conceptual framework for the study of food waste generation and prevention in the hospitality sector. Waste Manag. 49, 326–336. https://doi.org/10.1016/j.wasman.2016.01.017.
Papargyropoulou, E., Steinberger, J.K., Wright, N., Lozano, R., Padfield, R., Ujang, Z., 2019. Patterns and causes of food waste in the hospitality and food service sector: Food waste prevention insights from Malaysia. Sustainability 11. https://doi.org/10.3390/su11216016.
Schweizerische Eidgenossenschaft, 2022. Coronavirus: Bundesrat hebt Massnahmen auf – einzig Maskenpflicht im öffentlichen Verkehr und in Gesundheitseinrichtungen sowie Isolation bleiben noch bis Ende März [WWW Document]. URL https://www.admin.ch/gov/de/start/dokumentation/medienmitteilungen.msg-id-87216.html (accessed 9.13.24).
Scialabba, N., 2015. Food Wastage Footprint & Climate Change. FAO, Rome, Italy.
Silvennoinen, K., Heikkilä, L., Katajajuuri, J.-M., Reinikainen, A., 2015. Food waste volume and origin: Case studies in the Finnish food service sector. Waste Manag. 46, 140–145. https://doi.org/10.1016/j.wasman.2015.09.010.
Stöckli, S., Dorn, M., Liechti, S., 2018. Normative prompts reduce consumer food waste in restaurants. Waste Manag. 77, 532–536. https://doi.org/10.1016/j.wasman.2018.04.047.
Strotmann, C., Baur, V., Börnert, N., Gerwin, P., 2022. Generation and prevention of food waste in the German food service sector in the COVID-19 pandemic – Digital approaches to encounter the pandemic related crisis. Socio-Econ. Plann. Sci. 82. https://doi.org/10.1016/j.seps.2021.101104.
UNEP, 2024. Food Waste Index Report 2024. Think Eat Save: Tracking Progress to Halve Global Food Waste. United Nations Environment Programme, Nairobi.
United Nations, 2015. Transforming our world: the 2030 Agenda for Sustainable Development. United Nations, New York, NY, USA.
Vardopoulos, I., Abeliotis, K., Lasaridi, K., 2025. A systematic informetric analysis and literature review of food waste quantification studies in the food service industry. Sustainability 17, 103. https://doi.org/10.3390/su17010103.
Borstel, T. von, Prenzel, G.K., Welte, B., 2020. FOOD WASTE 4.0. Zwischenbilanz 2020. Reduktionsziele• Warenverluste• Umweltkennzahlen. Plankstadt. URL https://www.united-against-waste.de/downloads/united-against-waste-zwischenbilanz-2020-kompakt.pdf.
Wang, L.-E., Liu, G., Liu, X., Liu, Y., Gao, J., Zhou, B., Gao, S., Cheng, S., 2017. The weight of unfinished plate: A survey based characterization of restaurant food waste in Chinese cities. Waste Manag. 66, 3–12. https://doi.org/10.1016/j.wasman.2017.04.007.
Wang, F., Shreedhar, G., Galizzi, M.M., Mourato, S., 2022. A take-home message: workplace food waste interventions influence household pro-environmental behaviors. Resour., Conserv. Recycl. Adv. 15. https://doi.org/10.1016/j.rcradv.2022.200106.
Winnow Solutions Ltd, 2024. Case studies [WWW Document]. URL https://www.winnowsolutions.com/en/case-studies (accessed 9.10.24).
Wrap, 2013a. Household Food and Drink Waste in the United Kingdom 2012. WRAP, UK.
WRAP, 2013b. Overview of waste in the hospitality and food service sector [WWW Document]. URL https://www.wrap.ngo/resources/report/overview-waste-hospitality-and-food-service-sector (accessed 9.10.24).
Zornes, M., 2022. How IKEA co-workers teamed up with Winnow to halve their global food waste [WWW Document]. URL https://blog.winnowsolutions.com/ikea-winnow-halve-their-global-food-waste (accessed 9.10.24).